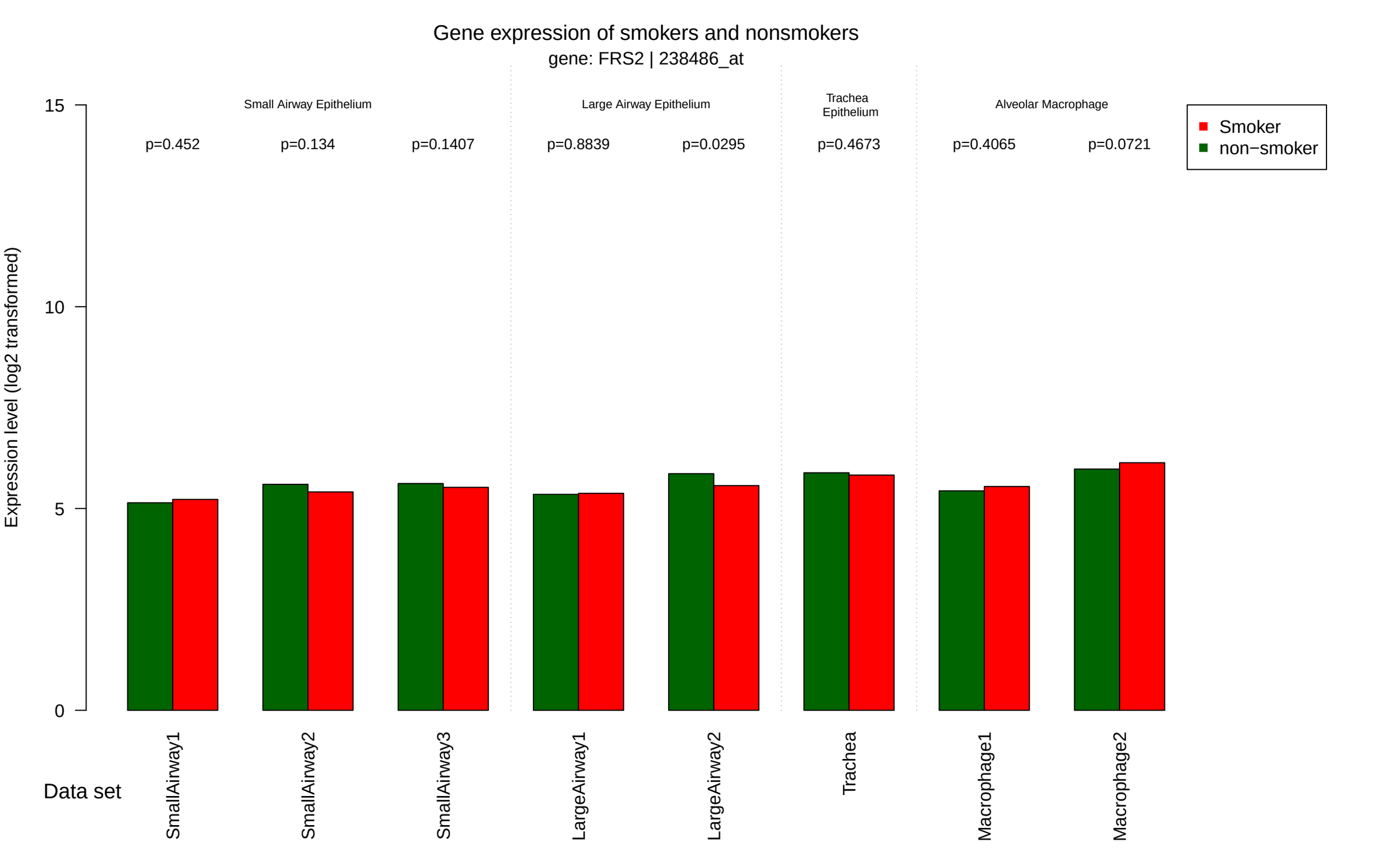
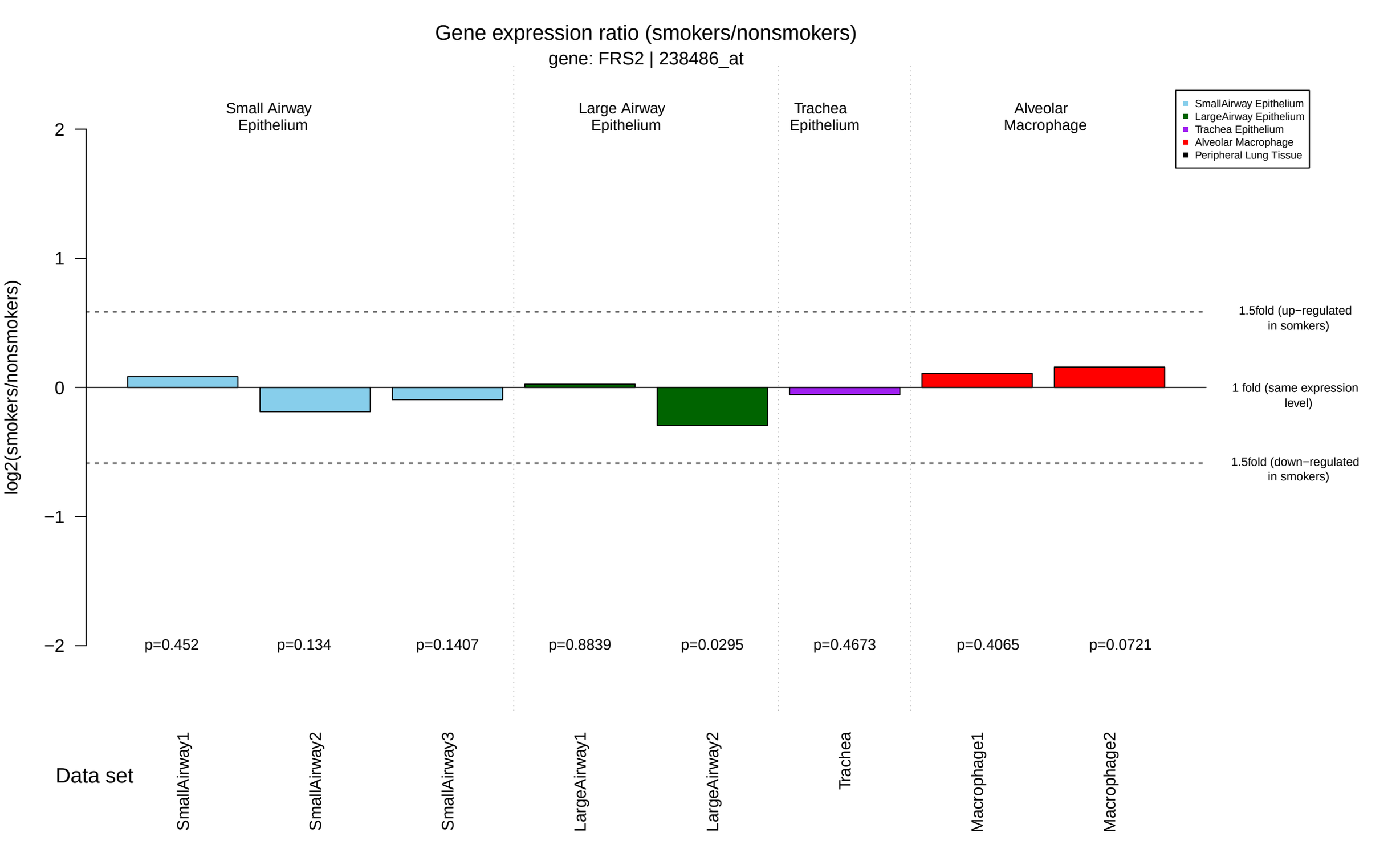
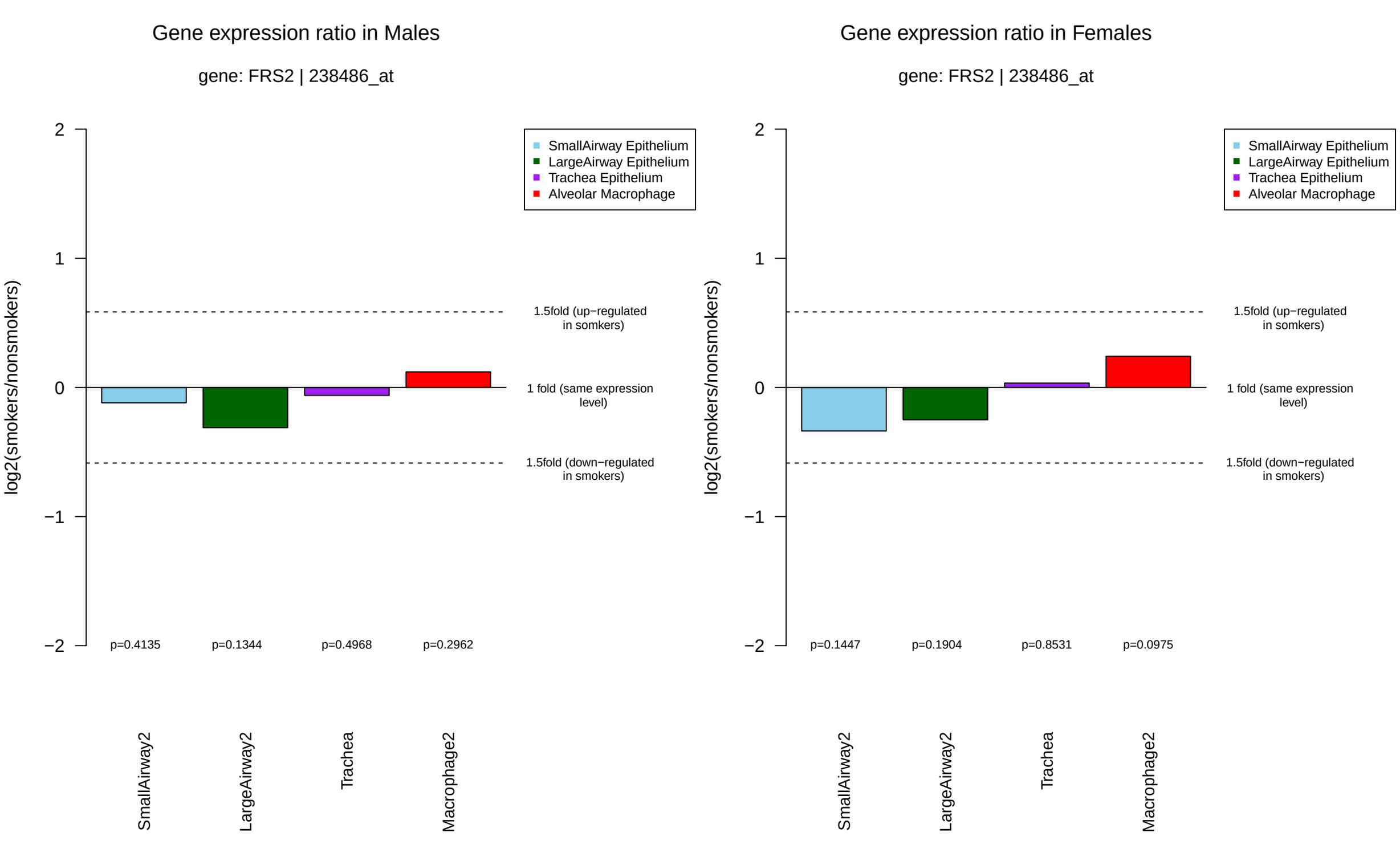
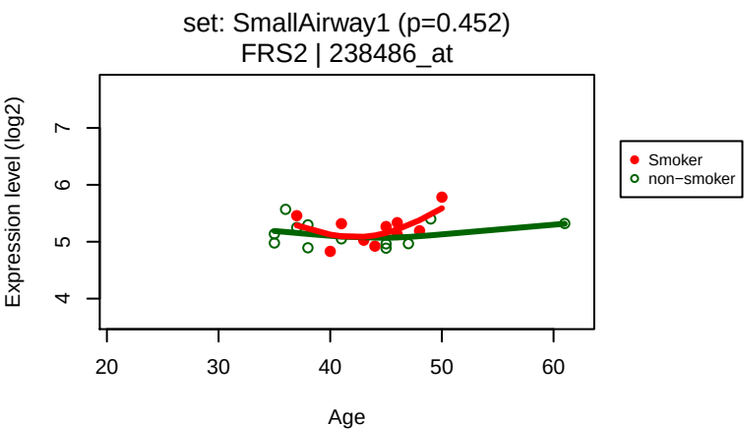
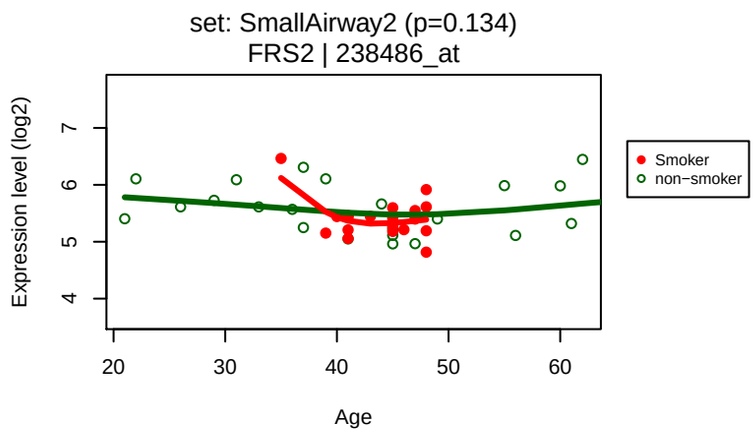
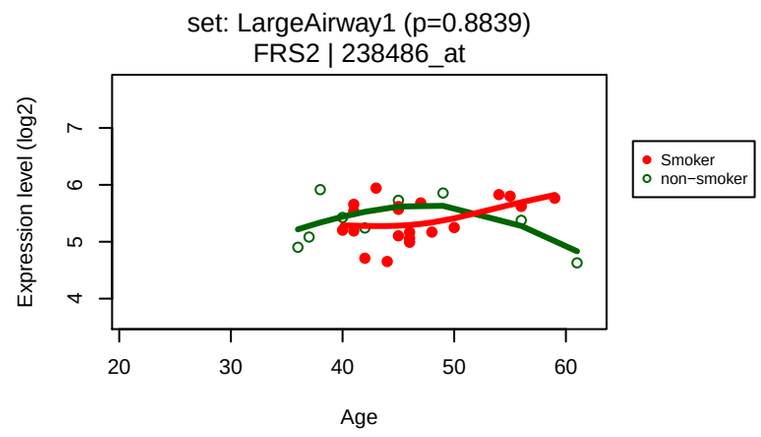

Gene expression of smokers and nonsmokers
gene: FRS2 | 238486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.452
p=0.134
p=0.1407
p=0.8839
p=0.0295
p=0.4673
p=0.4065
p=0.0721
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

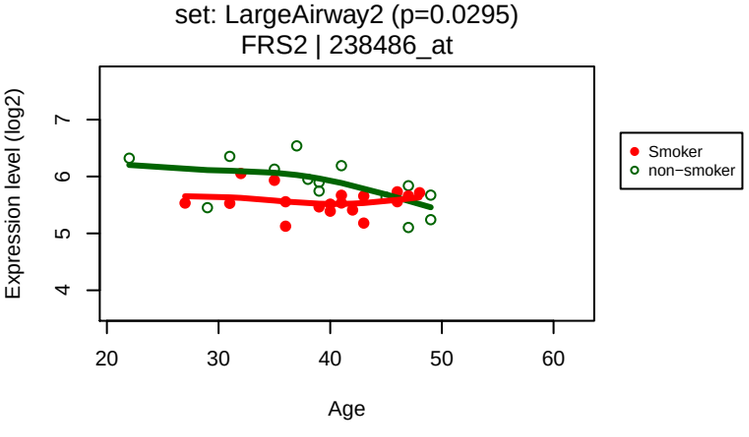

Gene expression ratio (smokers/nonsmokers)
gene: FRS2 | 238486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.452
p=0.134
p=0.1407
p=0.8839
p=0.0295
p=0.4673
p=0.4065
p=0.0721
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

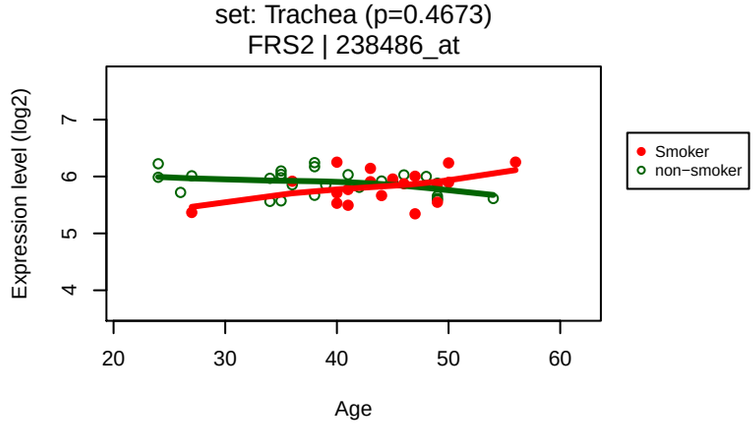

Gene expression ratio in Males
gene: FRS2 | 238486_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4135
p=0.1344
p=0.4968
p=0.2962
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRS2 | 238486_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1447
p=0.1904
p=0.8531
p=0.0975
SmallAirway2
LargeAirway2
Trachea
Macrophage2

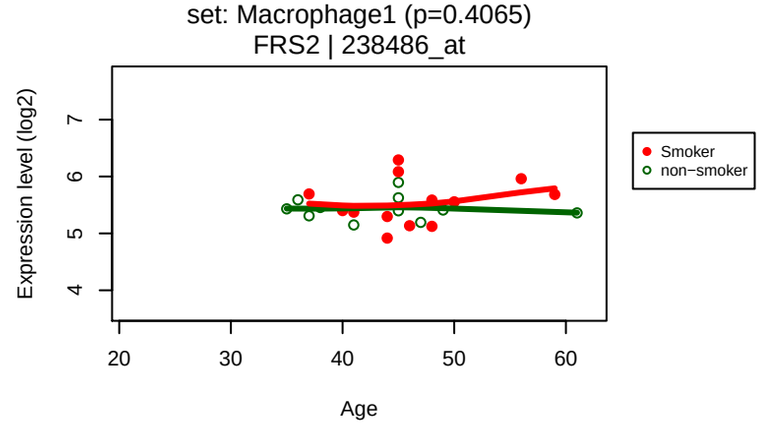

set: SmallAirway1 (p=0.452)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker

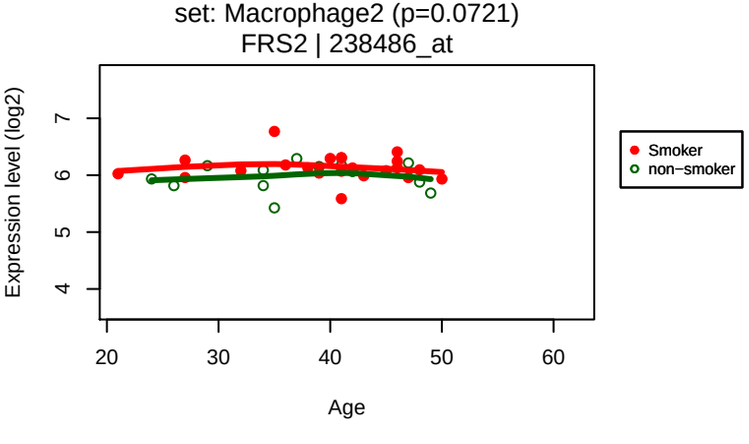

set: SmallAirway2 (p=0.134)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker

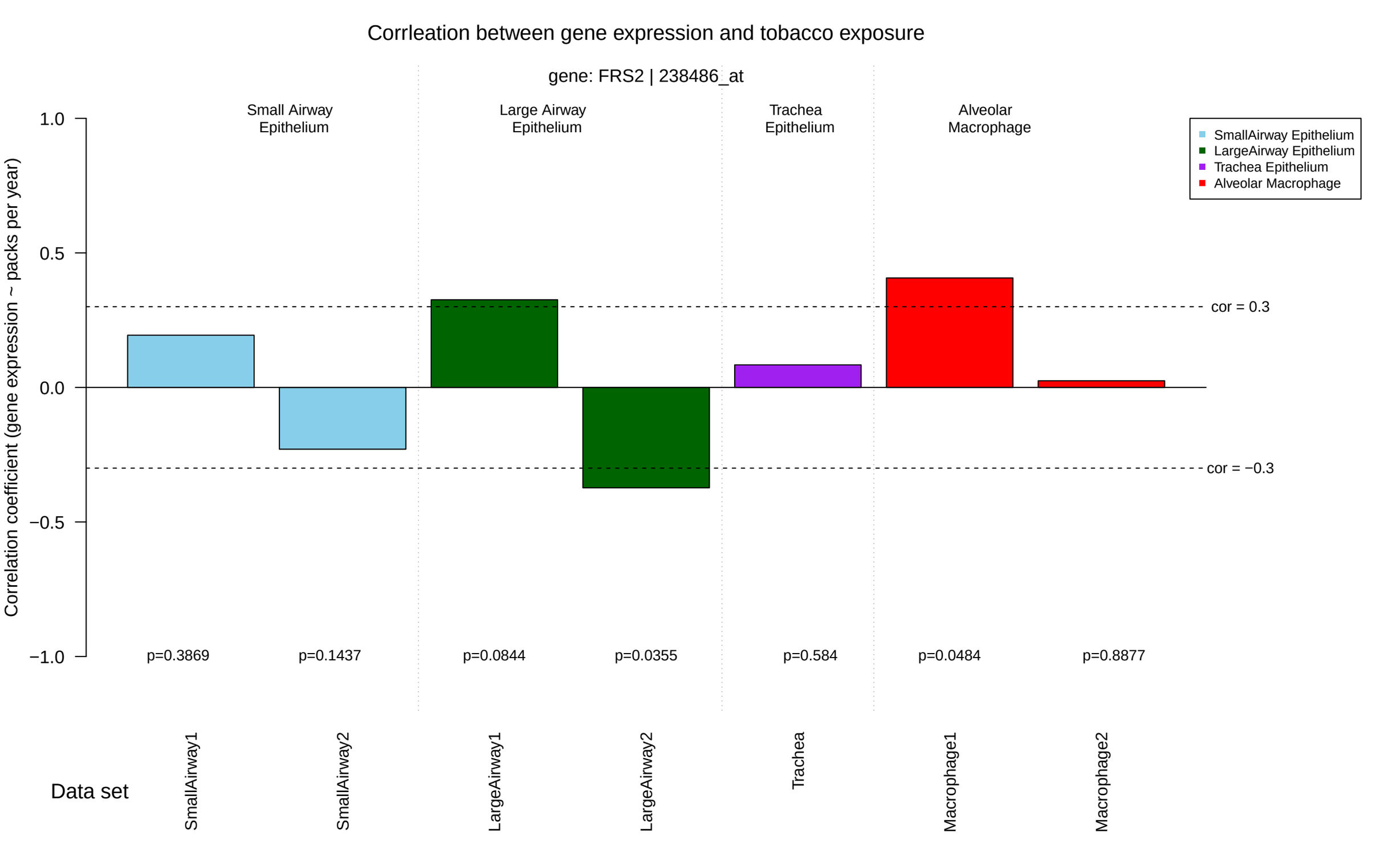

set: LargeAirway1 (p=0.8839)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0295)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4673)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4065)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0721)
FRS2 | 238486_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRS2 | 238486_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3869
p=0.1437
p=0.0844
p=0.0355
p=0.584
p=0.0484
p=0.8877
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2